```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.impute import KNNImputer, SimpleImputer
from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder,
LabelEncoder
from sklearn.compose import ColumnTransformer
from sklearn.naive_bayes import GaussianNB, CategoricalNB
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import classification_report
from mixed_naive_bayes import MixedNB

df = pd.read_csv("../J1/HeartDiseaseUCI.csv", index_col=0)

num_variables = ["age", "trestbps", "chol", "thalach", "ca",
"oldpeak"]
cat_variables = ["cp", "restecg", "thal"]
ord_variables = ["slope"]
bin_variables = ["sex", "fbs", "exang"]

df["target"] = np.where(df["num"] >= 1, 1, 0)

X = df.drop(["target", "num"], axis=1)
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size = 0.2,
    stratify = df["target"],
    random_state = 314
)

print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(242, 13)
(61, 13)
(242,)
(61,)
```

1. Gaussian Naive Bayes

Les prédictions ne sont faites que sur les variables numériques.

```python
num_pipeline = Pipeline(steps = [
    ("imputer", KNNImputer())
])

preprocessor = ColumnTransformer(transformers=
     [
        ("numeric", num_pipeline, num_variables),
     ],
     remainder="drop"
 )

pipeline = Pipeline(steps =[
    ("preprocessor", preprocessor),
    ("blabla", GaussianNB())
])

delta = 5*1e-10
var_smoothing_start = 1e-9 - delta
var_smoothing_end = 1e-9 + delta
var_smoothing_step = 1e-11

hyperparameters = {
    "blabla__var_smoothing" : np.arange(var_smoothing_start,
var_smoothing_end, var_smoothing_step)
}

gscv_gnb = GridSearchCV(
    estimator=pipeline,
    param_grid=hyperparameters,
    cv = 5,
    verbose = 4,
    scoring = "f1"
)

gscv_gnb.fit(X_train, y_train)
```

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Fitting 5 folds for each of 101 candidates, totalling 505 fits
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[CV 4/5] END blabla__var_smoothing=1.2299999999999976e-09;,
score=0.714 total time=   0.0s
[CV 5/5] END blabla__var_smoothing=1.2299999999999976e-09;,
score=0.800 total time=   0.0s
[CV 1/5] END blabla__var_smoothing=1.2399999999999974e-09;,
score=0.792 total time=   0.0s
[CV 2/5] END blabla__var_smoothing=1.2399999999999974e-09;,
score=0.714 total time=   0.0s
[CV 3/5] END blabla__var_smoothing=1.2399999999999974e-09;,
score=0.667 total time=   0.0s
[CV 4/5] END blabla__var_smoothing=1.2399999999999974e-09;,
score=0.714 total time=   0.0s
[CV 5/5] END blabla__var_smoothing=1.2399999999999974e-09;,
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score=0.800 total time=   0.0s
[CV 1/5] END blabla__var_smoothing=1.2499999999999975e-09;,
score=0.792 total time=   0.0s
[CV 2/5] END blabla__var_smoothing=1.2499999999999975e-09;,
score=0.714 total time=   0.0s
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score=0.714 total time=   0.0s
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score=0.714 total time=   0.0s
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[CV 4/5] END blabla__var_smoothing=1.2699999999999975e-09;,
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[CV 3/5] END blabla__var_smoothing=1.2899999999999974e-09;,
score=0.667 total time=   0.0s
[CV 4/5] END blabla__var_smoothing=1.2899999999999974e-09;,
score=0.714 total time=   0.0s
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```
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score=0.667 total time=   0.0s
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[CV 4/5] END blabla__var_smoothing=1.3299999999999973e-09;,
score=0.714 total time=   0.0s
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[CV 3/5] END blabla__var_smoothing=1.339999999999997e-09;, score=0.667
total time=   0.0s
[CV 4/5] END blabla__var_smoothing=1.339999999999997e-09;, score=0.714
```

```
total time=   0.0s
[CV 5/5] END blabla__var_smoothing=1.3399999999999997e-09;, score=0.800
total time=   0.0s
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[CV 3/5] END blabla__var_smoothing=1.3899999999999997e-09;, score=0.667
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[CV 4/5] END blabla__var_smoothing=1.3899999999999997e-09;, score=0.714
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score=0.792 total time=   0.0s
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score=0.714 total time=   0.0s
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score=0.667 total time=   0.0s
[CV 4/5] END blabla__var_smoothing=1.4399999999999967e-09;,
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score=0.714 total time=   0.0s
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[CV 3/5] END blabla__var_smoothing=1.4899999999999967e-09;,
score=0.667 total time=   0.0s
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score=0.714 total time=   0.0s
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score=0.800 total time=   0.0s

GridSearchCV(cv=5,
             estimator=Pipeline(steps=[('preprocessor',

ColumnTransformer(transformers=[('numeric',

Pipeline(steps=[('imputer',

KNNImputer())]),

['age',

'trestbps',

'chol',

'thalach',

'ca',

'oldpeak'])])),
                                       ('blabla', GaussianNB())]),
             param_grid={'blabla__var_smoothing': array([5.00e-10,
5.10e-10, 5.20e-10, 5.30e-10, 5.40e-10, 5.50e-10,
       5.60e-10, 5.70e-10, 5.80e-10...
       1.10e-09, 1.11e-09, 1.12e-09, 1.13e-09, 1.14e-09, 1.15e-09,
       1.16e-09, 1.17e-09, 1.18e-09, 1.19e-09, 1.20e-09, 1.21e-09,
       1.22e-09, 1.23e-09, 1.24e-09, 1.25e-09, 1.26e-09, 1.27e-09,
       1.28e-09, 1.29e-09, 1.30e-09, 1.31e-09, 1.32e-09, 1.33e-09,
       1.34e-09, 1.35e-09, 1.36e-09, 1.37e-09, 1.38e-09, 1.39e-09,
       1.40e-09, 1.41e-09, 1.42e-09, 1.43e-09, 1.44e-09, 1.45e-09,
       1.46e-09, 1.47e-09, 1.48e-09, 1.49e-09, 1.50e-09])},
             scoring='f1', verbose=4)
```

```
best_model = gscv_gnb.best_estimator_
best_hyperparameters = gscv_gnb.best_params_

best_hyperparameters

{'blabla__var_smoothing': 5e-10}

best_model

Pipeline(steps=[('preprocessor',
                 ColumnTransformer(transformers=[('numeric',

Pipeline(steps=[('imputer',

KNNImputer())]),
                                                  ['age', 'trestbps',
'chol',
                                                   'thalach', 'ca',
                                                   'oldpeak'])])),
                ('blabla', GaussianNB(var_smoothing=0.0))])

y_pred = best_model.predict(X_test)

print(classification_report(y_test, y_pred))

              precision    recall  f1-score   support

           0       0.69      0.82      0.75        33
           1       0.73      0.57      0.64        28

    accuracy                           0.70        61
   macro avg       0.71      0.69      0.70        61
weighted avg       0.71      0.70      0.70        61


print(classification_report(y_test, y_pred))

              precision    recall  f1-score   support

           0       0.69      0.82      0.75        33
           1       0.73      0.57      0.64        28

    accuracy                           0.70        61
   macro avg       0.71      0.69      0.70        61
weighted avg       0.71      0.70      0.70        61
```

2. Categorical Naive Bayes

Les prédictions ne sont faites que sur les variables catégorielles.

```
cat_pipeline = Pipeline(steps = [
    ("imputer", KNNImputer()),
    ("encoder", OneHotEncoder(drop = "first"))
])
ord_pipeline = Pipeline(steps = [
    ("imputer", KNNImputer()),
    ("encoder", OrdinalEncoder())
])
bin_pipeline = Pipeline(steps = [
    ("imputer", KNNImputer()),
])

preprocessor_2 = ColumnTransformer(transformers=
     [
        ("cat", cat_pipeline, cat_variables),
        ("ord", ord_pipeline, ord_variables),
        ("bin", bin_pipeline, bin_variables),
     ],
     remainder="drop"
 )

X_train
```

```
     age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang
oldpeak  \
96    52    1   4       128   255    0        0      161      1
0.0
114   43    0   4       132   341    1        2      136      1
3.0
68    54    1   3       150   232    0        2      165      0
1.6
229   54    1   4       110   206    0        2      108      1
0.0
80    58    1   4       150   270    0        2      111      1
0.8
..   ...  ...  ..       ...   ...  ...      ...      ...    ...      .
..
179   43    1   3       130   315    0        0      162      0
1.9
254   51    0   3       120   295    0        2      157      0
0.6
113   52    1   1       118   186    0        2      190      0
0.0
182   56    0   4       134   409    0        2      150      1
1.9
131   54    1   3       120   258    0        2      147      0
0.4

     slope   ca  thal
96       1  1.0   7.0
```

```
114      2  0.0   7.0
68       1  0.0   7.0
229      2  1.0   3.0
80       1  0.0   7.0
..     ...  ...   ...
179      1  1.0   3.0
254      1  0.0   3.0
113      2  0.0   6.0
182      2  2.0   7.0
131      2  0.0   7.0

[242 rows x 13 columns]
```

```
preprocessor_2
```

```
ColumnTransformer(transformers=[('cat',
                                 Pipeline(steps=[('imputer',
KNNImputer()),
                                                 ('encoder',

OneHotEncoder(drop='first'))]),
                                 ['cp', 'restecg', 'thal']),
                                ('ord',
                                 Pipeline(steps=[('imputer',
KNNImputer()),
                                                 ('encoder',
                                                  OrdinalEncoder())]),
                                 ['slope']),
                                ('bin',
                                 Pipeline(steps=[('imputer',
KNNImputer())]),
                                 ['sex', 'fbs', 'exang'])])
```

```
pipeline_2 = Pipeline(steps =[
    ("preprocessor", preprocessor_2),
    ("CategoricalNB", CategoricalNB())
])

pipeline_2.fit(X_train, y_train)
```

```
Pipeline(steps=[('preprocessor',
                 ColumnTransformer(transformers=[('cat',

Pipeline(steps=[('imputer',

KNNImputer()),

('encoder',

OneHotEncoder(drop='first'))]),
                                                  ['cp', 'restecg',
```

```
'thal']),
                                                  ('ord',

Pipeline(steps=[('imputer',

KNNImputer()),

('encoder',

OrdinalEncoder())]),
                                                   ['slope']),
                                                  ('bin',

Pipeline(steps=[('imputer',

KNNImputer())]),
                                                   ['sex', 'fbs',
'exang'])])),
                ('CategoricalNB', CategoricalNB())])
```

```
y_pred = pipeline_2.predict(X_test)

print(classification_report(y_pred, y_test))
```

```
              precision    recall  f1-score   support

           0       0.91      0.86      0.88        35
           1       0.82      0.88      0.85        26

    accuracy                           0.87        61
   macro avg       0.87      0.87      0.87        61
weighted avg       0.87      0.87      0.87        61
```

3. Mixed Naive Bayes

Les données catégorielles doivent être encodées avec un LabelEncoder. Mais il y a un problème de compatibilité avec les Pipelines lors du prétraitement.

On impute les valeurs, puis on LabelEncode chaque colonne catégorielle avec un LabelEncoder, mais sans Pipeline.

Attention au ColumnTransformer qui change l'ordre des colonnes.

```
X_train_clean = X_train.copy()
X_test_clean = X_test.copy()

# Imputation des variables numériques
imputer = KNNImputer()
X_train_clean[num_variables] =
imputer.fit_transform(X_train_clean[num_variables])
```

```python
X_test_clean[num_variables] =
imputer.transform(X_test_clean[num_variables])

# Imputation des variables catégorielles
columns_cat = cat_variables + ord_variables + bin_variables

imputer = SimpleImputer(strategy="most_frequent")
X_train_clean[columns_cat] =
imputer.fit_transform(X_train_clean[columns_cat])
X_test_clean[columns_cat] =
imputer.transform(X_test_clean[columns_cat])

# Encodage des variables catégorielles (LabelEncoder)
le = LabelEncoder()
for col in columns_cat:
    X_train_clean[col] = le.fit_transform(X_train_clean[col])
    X_test_clean[col] = le.transform(X_test_clean[col])

# Indices des colonnes catégorielles
indices = [list(X_train_clean.columns).index(c) for c in columns_cat]

# Création du modèle MixedNB
model_mixed_nb = MixedNB(categorical_features = indices)

model_mixed_nb.fit(X_train_clean, y_train)
```

```
MixedNB(alpha=0.5, var_smoothing=1e-09)
```

```python
y_pred = model_mixed_nb.predict(X_test_clean)

print(classification_report(y_pred, y_test))
```

```
              precision    recall  f1-score   support

           0       0.85      0.80      0.82        35
           1       0.75      0.81      0.78        26

    accuracy                           0.80        61
   macro avg       0.80      0.80      0.80        61
weighted avg       0.81      0.80      0.80        61
```

Votre Jupyter Notebook présente une implémentation complète des modèles Naive Bayes pour la classification, en utilisant différentes variantes adaptées aux types de données. Vous avez également exploré l'utilisation d'un modèle Mixed Naive Bayes pour gérer simultanément des données numériques et catégorielles. Voici quelques conseils et corrections pour optimiser votre code et améliorer la clarté et la performance des modèles.

1. Gaussian Naive Bayes

- Votre utilisation de `KNNImputer` pour les variables numériques est appropriée, surtout s'il y a des valeurs manquantes significatives.

- L'utilisation d'un intervalle très étroit pour `var_smoothing` est intéressante et peut aider à affiner le modèle, mais assurez-vous que cette granularité est nécessaire étant donné le coût computationnel potentiel.
- Vous avez utilisé deux fois `print(classification_report(y_test, y_pred))`. La suppression de l'une des deux peut aider à éviter la redondance.

2. **Categorical Naive Bayes**

- Votre traitement des variables catégorielles semble correct. Cependant, utiliser `KNNImputer` pour les variables catégorielles n'est pas idéal, car ce type d'imputeur est conçu pour les variables continues. `SimpleImputer` avec la stratégie "most_frequent" serait plus approprié ici.
- Votre pipeline et l'encodage sont bien configurés. Assurez-vous que l'encodage "drop first" est compatible avec votre analyse.

3. **Mixed Naive Bayes**

- L'utilisation de `MixedNB` est une bonne stratégie pour gérer à la fois les variables catégorielles et numériques.
- Votre approche pour contourner le problème de compatibilité avec le pipeline en effectuant manuellement l'imputation et l'encodage est compréhensible. Cependant, c'est moins idéal en termes de réplicabilité et de risque d'erreur humaine. Évaluer l'utilisation de `FunctionTransformer` ou d'autres méthodes pour intégrer ces étapes dans un pipeline pourrait être une option à long terme.
- Assurez-vous que l'utilisation du `LabelEncoder` est correcte ici, car normalement il est utilisé pour coder la variable cible. Pour les variables indépendantes dans un contexte multiclasse, `OrdinalEncoder` est souvent plus approprié.

Conseils Généraux

- **Visualisation des Performances**: Inclure des courbes ROC pour chaque modèle pourrait fournir une meilleure compréhension des performances.
- **Validation Croisée**: Pour améliorer l'évaluation des modèles, envisagez d'utiliser une validation croisée plus robuste si ce n'est pas déjà le cas.
- **Nettoyage du Code**: Assurez-vous que le code est bien organisé et documenté pour faciliter la compréhension et la maintenance. Commentez chaque étape importante pour clarifier l'objectif et les méthodes utilisées.

Amélioration de la Documentation

- **Explications des Choix**: Documentez pourquoi certains choix ont été faits, par exemple, l'usage spécifique de certaines méthodes d'imputation ou d'encodage.
- **Réflexions sur les Résultats**: Après chaque classification, ajoutez une discussion ou une analyse des résultats obtenus, expliquant ce qui pourrait être amélioré ou ce qui a bien fonctionné.

Votre exploration de différentes techniques de Naive Bayes pour traiter des types de données variés est bien menée. Continuez à expérimenter et à peaufiner vos modèles pour maximiser leur efficacité.